

Secure FFT IP using C way Partitioning based Obfuscation and Fingerprint

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INTRODUCTION

- The FFT is a fundamental building block used in signal processing systems, with applications ranging from Digital MODEMs, to ultrasound and Computed Tomography (CT) image reconstruction.
- However, no work focused on designing its secure IP design using HLS..

RELATED WORK

Sr. No.	Existing Work	Technique Used	Remarks
1.	A. Sengupta et. al., [6] (2019)	Steganography offers security through embedded stego constraints.	However, none of these approaches [6], [7] are robust as they can be compromised by an adversary by decoding auxiliary security parameters. Moreover, these approaches are not capable of hindering an attacker to perform RTL alteration, as they do not have provision for HLS-based RTL structural obfuscation, unlike proposed approach
2.	F. Koushanfar et. al., [7] (2005)	Multi-variable signature combinations for security	

A. Threat Model and Motivation of Proposed Methodology

- In this paper, this threat is handled by employing C-way partitioning-based structural RTL obfuscation by an IP vendor in order to hide the functionality and structural interconnection of IP design.
- The effects of RTL obfuscation on layout obfuscation have been established in several literatures [2], [13]. Further as established in [6], [7], [8], an adversary in the SoC house, may attempt to perform IP piracy and falsely claim of IP.
- Therefore, these threats are tackled using proposed IP vendor fingerprint biometric as detective countermeasure.

PROPOSED WORK

A. Overview

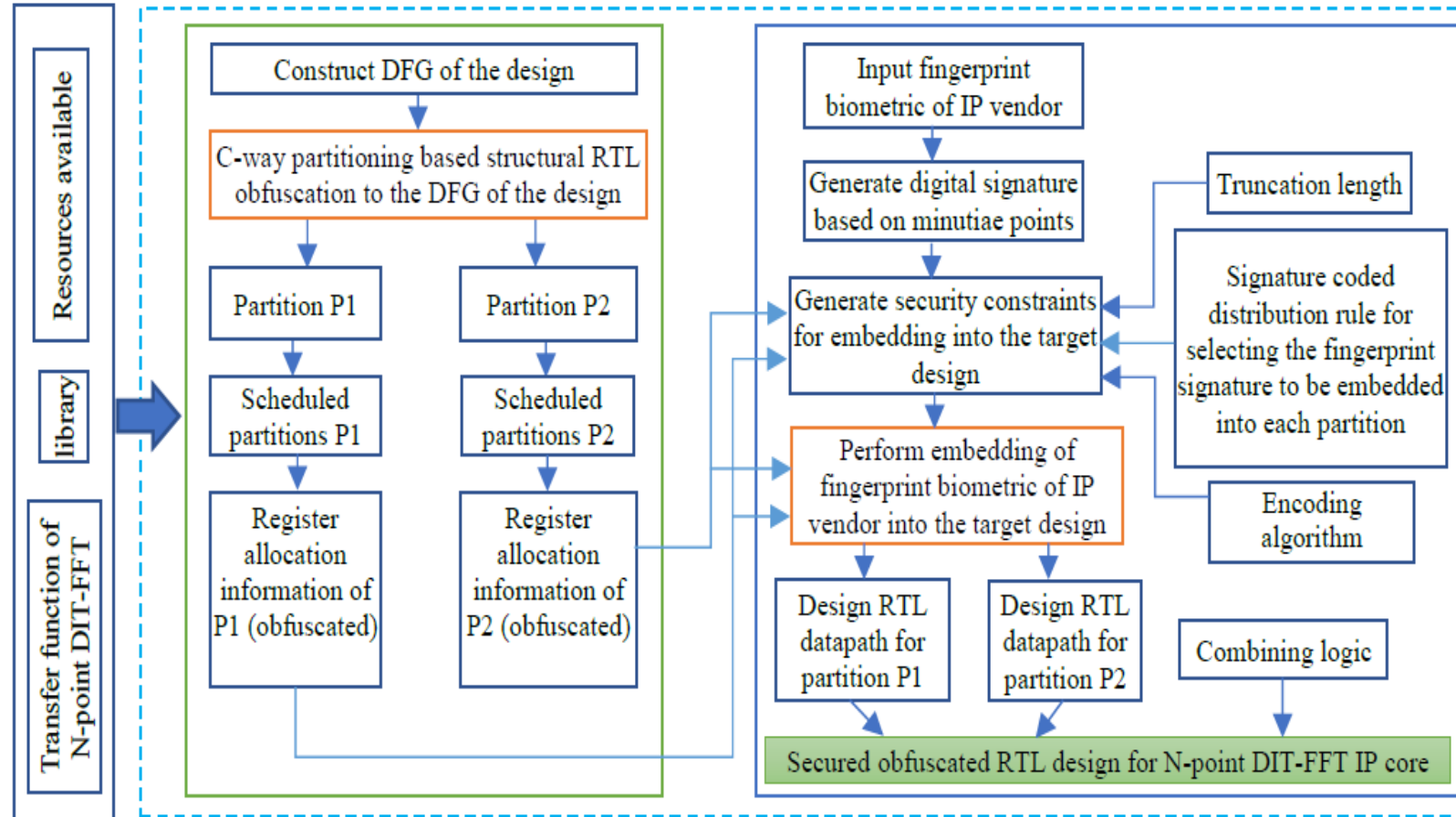


Fig. 1. Design flow of the proposed security methodology

B. Secured Scheduled FFT Design using Proposed Approach

- Transfer function of Discrete Fourier Transform (DFT) is represented as [8]:

$$X(k) = \sum x(n) W_N^{nk}, 0 \leq K \leq N - 1 \quad (1)$$

$$X(k) = G(k) + W_N^k H(k), k = 0, 1, \dots, \frac{N}{2} - 1 \quad (2)$$

$$PC_S = \left(1 - \frac{1}{R_B}\right)^x \quad (6)$$

PROPOSED WORK

B. Secured Scheduled FFT Design using Proposed Approach

Now based on (2), the derived signal flow graph for $N=8$ has the following representations:

$$X(0) = G(0) + W^{80} * H(0)$$

$$X(1) = G(1) + W^{81} * H(1)$$

$$X(2) = G(2) + W^{82} * H(2)$$

$$X(3) = G(3) + W^{83} * H(3)$$

$$X(4) = G(0) + W^{84} * H(0)$$

$$X(5) = G(1) + W^{85} * H(1)$$

$$X(6) = G(2) + W^{86} * H(2)$$

$$X(7) = G(3) + W^{87} * H(3)$$

PROPOSED WORK

Converting the transfer function into C-way partitioned structurally obfuscated scheduled design—

- The above equations are converted into CDFG, shown in Fig. 2 (a). An IP vendor can perform $Q=2^{(\log_2 N)-1}$ types of partitioning depending on the number of data elements in each partition.

# Applied partitions by IP vendor	# Elements in each partition	Resultant partition examples
C=7	V1=1, V2=1,.....,V7=1	P1[x(0)]; P2[x(1)];P3[x(2)];P4[x(3)];.....P7[x(7)]
C=4	V1=2, V2=2,.....,V4=2	P1[x(0)x(1)]; P2[x(2),(3)];P3[x(4)x(5)];P4[x(6)x(7)]
C=3	V1=3,V2=3,V3=2	P1[x(0)x(1)x(2)]; P2[x(3)x(4)x(5)];P3[x(6)x(7)]
C=2	V1=4, V2=4	P1[x(0)x(1)x(2)x(3)]; P2[x(4)x(5)x(6)x(7)]
C=2	V1=5, V2=3	P1[x(0)x(1)x(2)x(3)x(4)]; P2[x(5)x(6)x(7)]
C=2	V1=6, V2=2	P1[x(0)x(1)x(2)x(3)x(4)x(5)]; P2[x(6)x(7)]
C=2	V1=7, V2=1	P1[x(0)x(1)x(2)x(3)x(4)x(5)x(6)]; P2[x(7)]

Table 1. Example of resulting partitions for $Q=7$

PROPOSED WORK

- Converting the transfer function into C-way partitioned structurally obfuscated scheduled design—

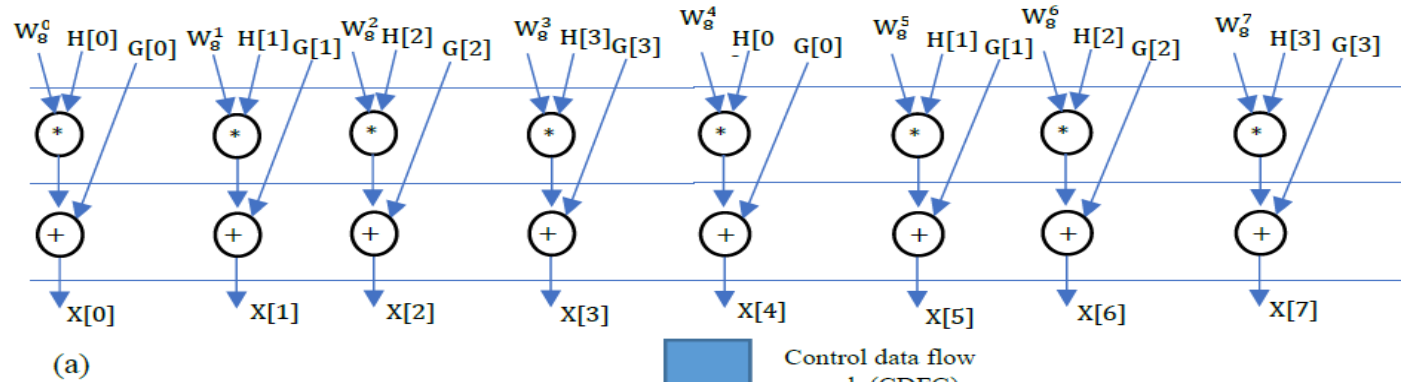


Fig. 2. (a) Data flow graph corresponding to DIT-FFT IP core for $N=8$

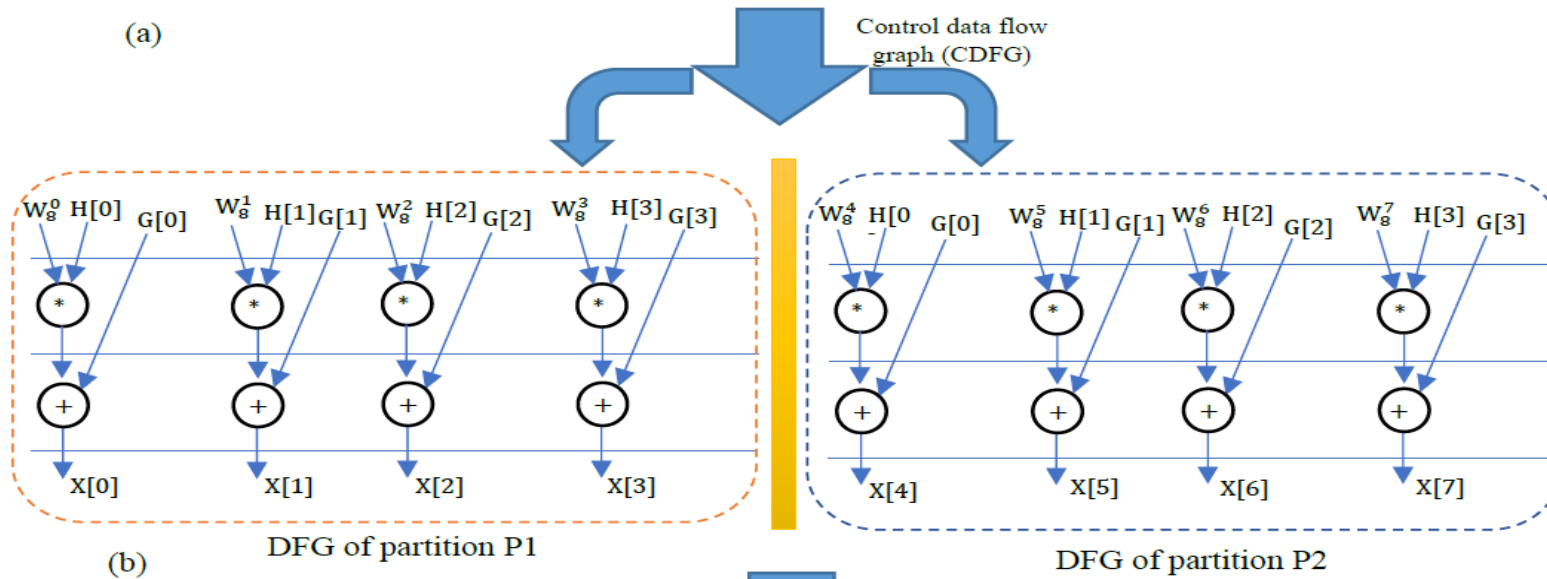


Fig. 2. (b) C- way partitioned DFG of FFT for $C=2$, $V_1=V_2=4$ and $N=8$

PROPOSED WORK

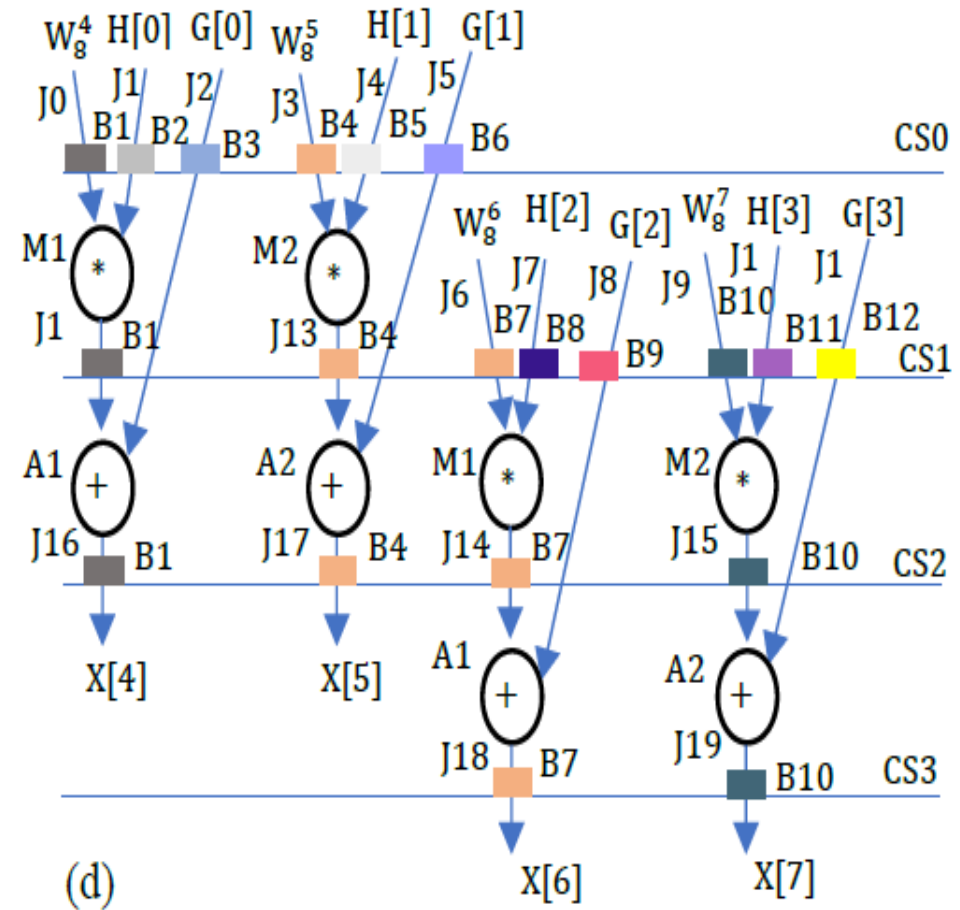
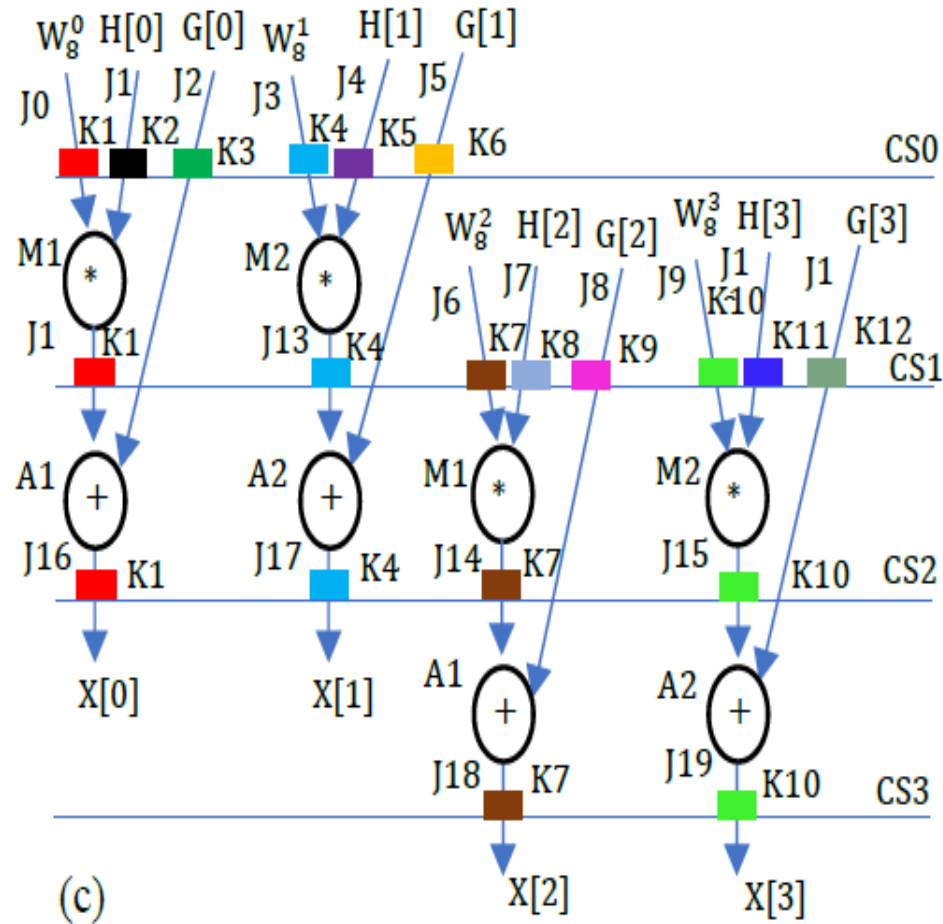


Fig. 2. (c) Scheduled DFG of structurally obfuscated partition P1 for 2A, 2M indicating storage variables (J0 to J19) and register (K1 to K12) allocation. **(d)** Scheduled DFG of structurally obfuscated partition P2 for 2A, 2M indicating storage variables and register (B1 to B12) allocation.

C. Generating the Fingerprint Biometric Digital Template

The following subprocesses are executed (Fig. 3):

Sub-process-1: capture the fingerprint biometric of IP vendor using a scanning device.

Subprocess-2: preprocess the obtained image and perform the image enhancement using FFT in the obtained image to reconnect the broken ridges. Subsequently, perform the binarization and thinning process for smooth and accurate extraction of fingerprint minutiae points.

Subprocess-3: operate the thinned fingerprint image to extract the minutiae points features.

Subprocess-4: transform each of the minutiae points into their binarized form.

PROPOSED WORK

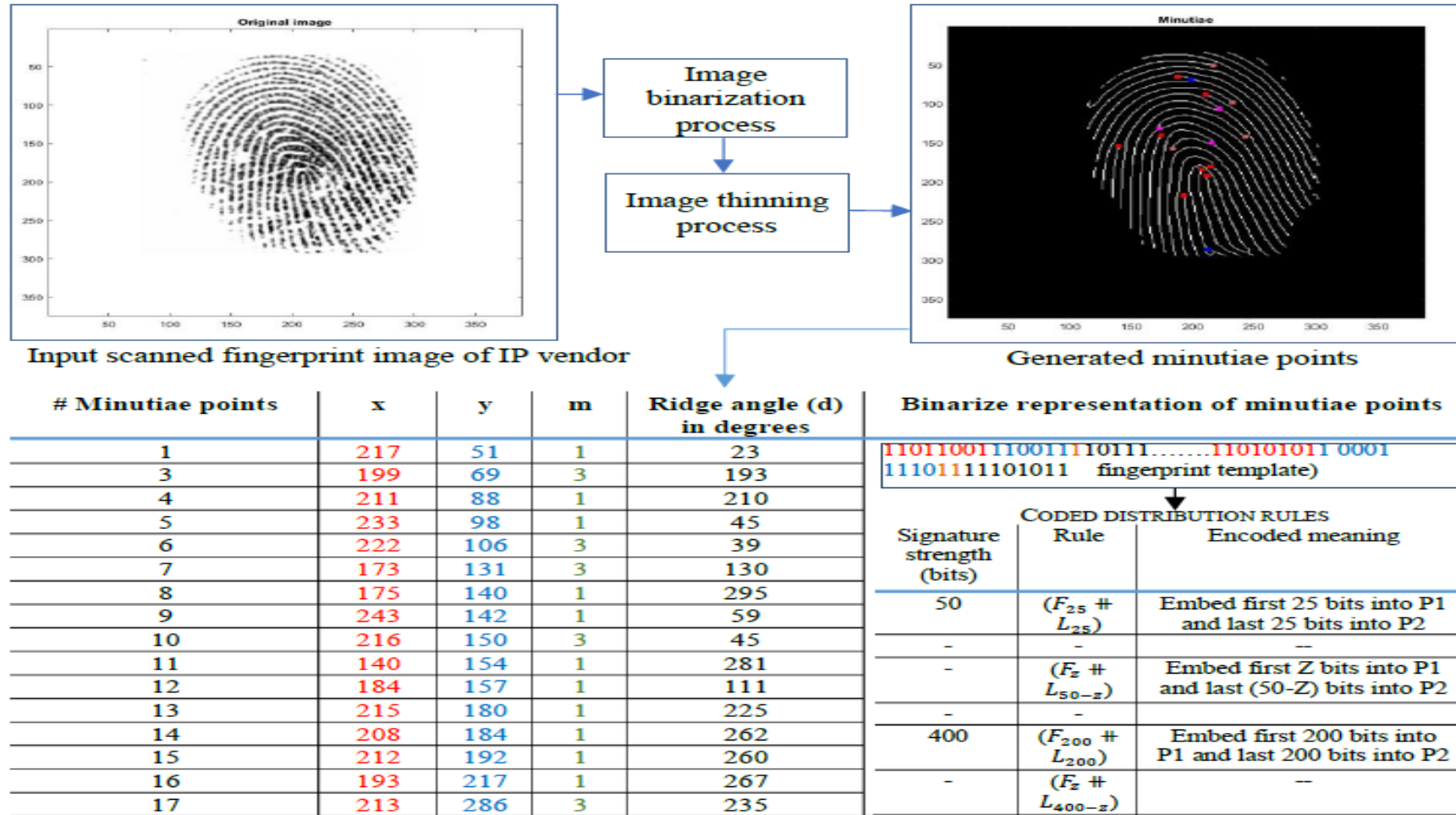


Fig. 3. Generation of coded fingerprint signature

D. Generating the Hardware Security Constraints from Obfuscated Scheduled Design and Fingerprint Biometric .

For the sake of brevity, we are considering the fingerprint signature strength of 100 bits (including first 50 and last 50 bits of fingerprint signature). The first 50 bits of the generated fingerprint signature are:

“11011001-110011-1-10111-10111100-1000001-1-101100101-11000”.

D. Generating the Hardware Security Constraints from Obfuscated Scheduled Design and Fingerprint Biometric

IP Vendor's Encoding rule:

Rule-1: generate a security constraint by forming the storage variable pair (even-even) corresponding to signature bit '0'.

Rule-2: generate a security constraint by forming the storage variable pair (odd-odd) corresponding to signature bit '1'.

PROPOSED WORK

E. Generating the Secured RTL Datapath for N-point DIT-FFT Design with Embedded Fingerprint Biometric

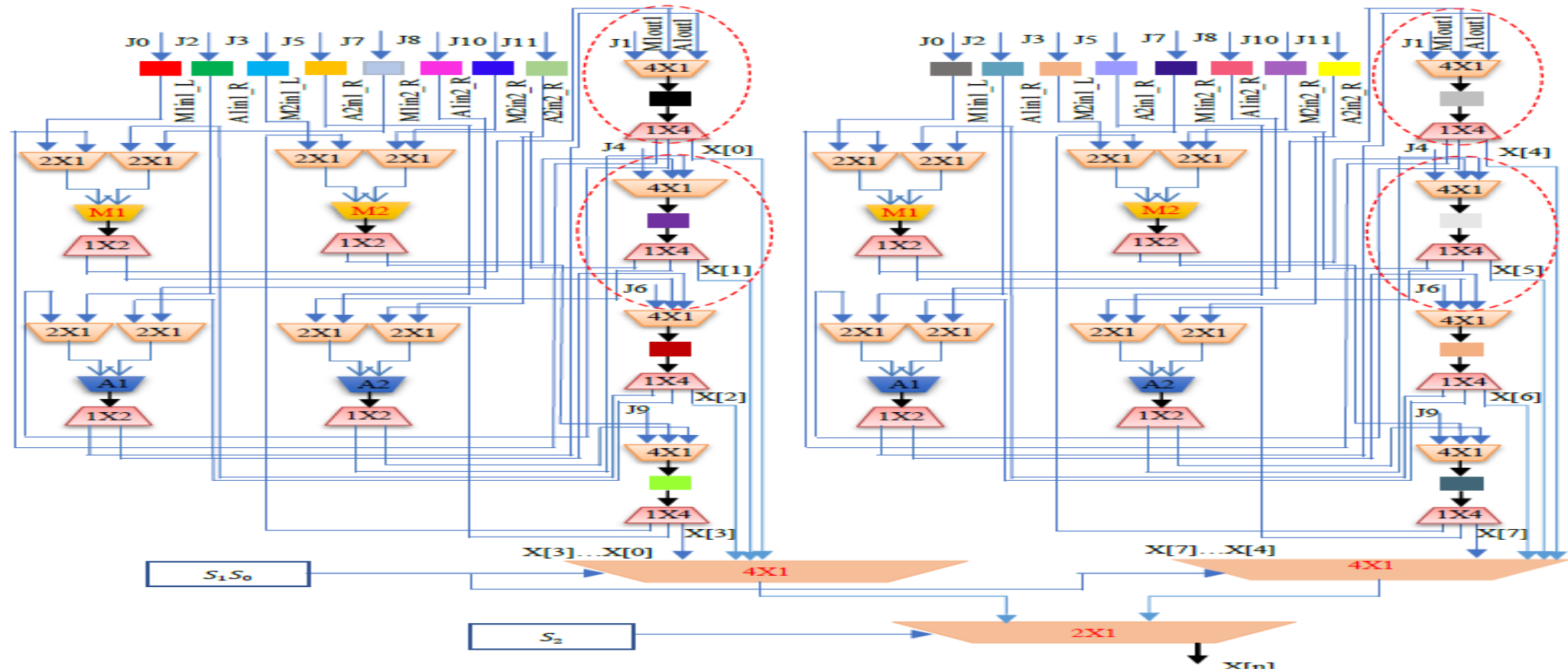


Fig. 4. Secured RTL Datapath design of N-point DIT-FFT (for N=8)

PROPOSED WORK

E. Generating the Secured RTL Datapath for N-point DIT-FFT Design with Embedded Fingerprint Biometric

Table 1. Locally altered register allocation information (post embedding first fifty bits of fingerprint signature into partition-P1).

Registers	K1	K2	K3	K4	K5	K6	K7	K8	K9	K10	K11	K12
CS0	J0	J1	J2	J3	J4	J5	--	--	--	--	--	--
CS1	J12	J12	-	J13	J13	--	J6	J7	J8	J9	J10	J11
CS2	J16	J16	-	J17	J17	--	J14	--	--	J15	--	--
CS3							J18			J19		

IMPLEMENTATION RESULTS

A. Analysis of Security due to Structural Obfuscation

- The hardness of structural obfuscation has been evaluated in [1], [5] and [11], using standard ‘power of obfuscation’ metric. ‘Power of obfuscation (Pobf)’ is expressed using following equations indicating normalized value between 0 to 1:’

$$P^{obf} = \frac{n}{n^T} \quad (3)$$

- Lower the probability, harder it is for an adversary to combine the partitions to the retrieve IP RTL design, during RE attempt. It can be represented as:

$$P_T = \frac{1}{2^{(\log_2 N) - 1}} \times \frac{1}{C} \quad (4)$$

Table II.A reports the P^{obf} and P_T of the proposed approach

IMPLEMENTATION RESULTS

A. Analysis of Security due to Structural Obfuscation

Table II. The P^{obf} and P_T of the proposed approach

N=8 (8-point FFT)			N=16 (16-point FFT)			N=32 (32-point FFT)		
C	P^{obf}	P_T	C	P^{obf}	P_T	C	P^{obf}	P_T
2	0.125	0.07	2	0.062	0.03	4	0.062	0.008
3	0.187	0.04	4	0.125	0.01	8	0.125	0.004
4	0.25	0.03	8	0.25	0.008	15	0.234	0.002
7	0.437	0.02	12	0.375	0.005	24	0.375	0.001
8	0.5	0.01	16	0.5	0.004	48	0.75	0.0006

IMPLEMENTATION RESULTS

A. Analysis of Security due to Structural Obfuscation

- Further, the attacker's effort needed in terms of de-obfuscating the gate level IP design is measured in terms of probability of identifying all obfuscated gates using brute force.
- The probability of identifying all obfuscated gates, due to C-way partitioning, from an attacker's perspective during applying brute force attack is given as:

$$P_G^T = \pi_{i=1}^{SC} \left(\frac{1}{|G_S^T|} \right) \quad (5)$$

- The affected gates (SC) due to proposed C-way partitioning resulting into structural obfuscation and P_G^T is presented in Table II.B.

IMPLEMENTATION RESULTS

A. Analysis of Security due to Structural Obfuscation

Table II.B. The affected gates (SC) due to proposed C-way partitioning resulting into structural obfuscation and P_G^T

Finger print Signature Size	Un – <i>obfuscated</i> <i>design gate</i> <i>count</i> (G_U^T)	Proposed Secured design gate count	Affected gates due to proposed C-way partitioning $SC =$ $ G_S^T - G_U^T $	Attacker's effort in de-obfuscating the proposed IP design (P_g^T)
		Post obfuscation without with Fingerprint (G_S^T)		
50	4288	9502 9502	5214	$1 \div (9502)^{5214}$
100	4288	9502 9502	5214	$1 \div (9502)^{5214}$
150	4288	9502 9502	5214	$1 \div (9502)^{5214}$
200	4288	9502 9502	5214	$1 \div (9502)^{5214}$
400	4288	9502 9502	5214	$1 \div (9502)^{5214}$

IMPLEMENTATION RESULTS

B. Analysis of Security due to Fingerprint

- Probability of Coincidence :

$$PCs = \left(1 - \frac{1}{R_B}\right)^x \quad (6)$$

Where 'R_B' and 'x' designate the count of storage elements in baseline design and number of generated security constraints.

- Tamper tolerance :

$$TT_A = V^x \quad (7)$$

Where, 'V' represents the number of distinct bits of the signature.

C. Entropy Estimation of the Proposed Approach

➤ The entropy of the proposed approach can be determined as follows:

$$E_t = \frac{1}{n!} \times \frac{1}{m!} \times \frac{1}{2^x} \quad (8)$$

*Where 'n' indicates the number of features of each minutia,
'm' indicates total number of minutiae points of fingerprint,
'x' indicates strength of signature and
'2^x' indicates total signature space.*

IMPLEMENTATION RESULTS

TABLE III

COMPARISON OF PC_S , TT_A AND E_t ACHIEVED USING THE PROPOSED SECURITY METHODOLOGY W.R.T [6], [7]

Proposed approach					Hardware watermarking [7] (For $x=240$ bits)				IP steganography [6] (For $x=125$ bits)				DNA based watermarking [10] (For $x=128$ bits)			
x	PC_S	TT_A	E_t	%Ov.	PC_S	TT_A	E_t	%Ov.	PC_S	TT_A	E_t	%Ov.	PC_S	TT_A	E_t	%Ov.
250	2.3 E-5	1.8 E+75	6.4 E-92	0.0%	3.66 E-5	1.7 E+72	5.6 E-73	0.0%	4.8 E-3	N/A	2.3 E-38	0.0%	4.8 E-3	3.4 E+38	8.63 E-78	0.0%
300	2.8 E-6	2.0 E+90	5.7 E-107	0.0%	3.66 E-5	1.7 E+72	5.6 E-73	0.0%	4.8 E-3	N/A	2.3 E-38	0.0%	4.8 E-3	3.4 E+38	8.63 E-78	0.0%
350	3.3 E-7	2.2 E+105	5.1 E-122	0.0%	3.66 E-5	1.7 E+72	5.6 E-73	0.0%	4.8 E-3	N/A	2.3 E-38	0.0%	4.8 E-3	3.4 E+38	8.63 E-78	0.0%
400	4.0 E-8	2.5 E+120	4.5 E-137	0.0%	3.66 E-5	1.7 E+72	5.6 E-73	0.0%	4.8 E-3	N/A	2.3 E-38	0.0%	4.8 E-3	3.4 E+38	8.63 E-78	0.0%

IMPLEMENTATION RESULTS

D. Detection of IP Piracy

- In the proposed approach, secret security constraints of biometric signature are extracted from the suspected chip design and matched with the originally embedded biometric security constraints of the IP design.
- From the extracted layout design file of the suspected chip, through backward engineering, the IP core RTL is obtained.
- Finally, the secret security constraints of biometric signature are extracted from the IP RTL file for matching [7], [10].
- In case of complete matching, IP piracy is detected.

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Thank You!